PROJECT REPORT
RAG-based Question Answering system

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Project Overview: An advanced AI system built to accurately answer user questions based strictly on the Swiggy Annual Report 2023-24. This project implements a sophisticated Retrieval-Augmented Generation (RAG) pipeline designed to prevent hallucinations and provide context-grounded financial insights.

The objective of this system is to allow users to ask natural language questions regarding Swiggy's business operations and receive accurate answers derived solely from the official document.

(Github Repo)

Tech Stack:

- Programming Language: Python
- Framework: Langchain
- Vector Database: ChromaDB
- Embedding Model: SentenceTransformers
- LLM: Llama 3.2 3B Instruct
- UI: Streamlit
- Evaluation: Ragas Framework

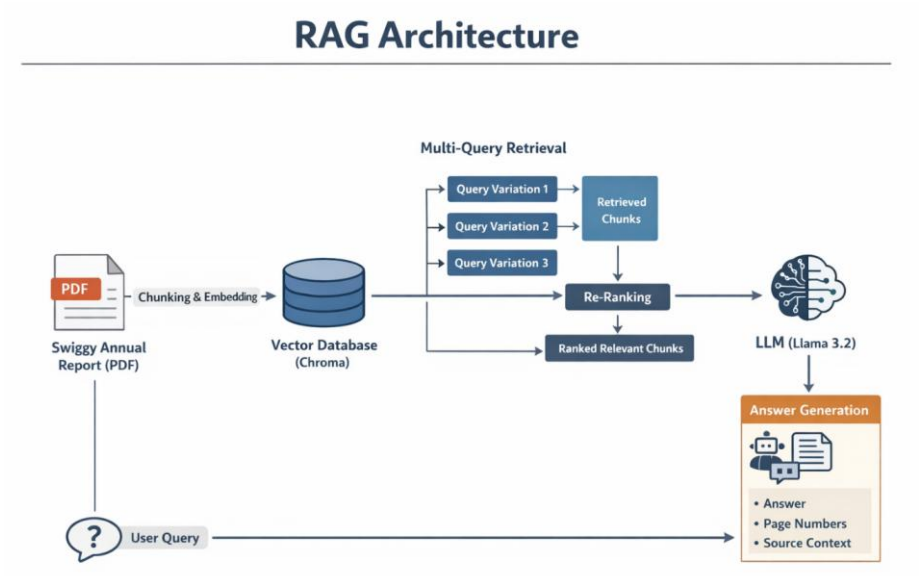
Dataset:

Document: Swiggy Annual Report (Latest Public Version)
Format: PDF
The document serves as the **only knowledge source** for the RAG pipeline.

System Architecture:

Pipeline Overview:

PDF → Chunking → Embeddings → Vector Database → Query Input
→ Multi-Query Retrieval → Re-Ranking → LLM Answer Generation
→ Answer + Page Citations + Context



Implementation Overview:

- **Document Processing:** The PDF is loaded using **PyPDFLoader** and split into semantic chunks using RecursiveCharacterTextSplitter. Chunk configuration: (Chunk Size: 1000, Chunk Overlap: 200)
  Metadata such as page numbers is preserved for citations.

- **Embeddings & Vector Store:** Text chunks are converted into embeddings using-**SentenceTransformer.** Embeddings are stored in a **Chroma Vector Database**, enabling semantic similarity search.

- **Retrieval-Augmented Generation:** Relevant chunks are retrieved from the vector database and passed to the LLM to generate answers grounded in the document context.
  LLM used: **Llama-3.2-3B-Instruct**

Key Features Implemented

- **Document Processing:**
  - PDF loading and text chunking with metadata.
- **Embeddings & Vector Store:**
  - Semantic embeddings stored in Chroma vector database.
- **Retrieval-Augmented Generation:**
  - Relevant document chunks retrieved and used for answer generation.
- **Question Answering Interface:**
  - Interactive **Streamlit chat interface** for user queries.
  - Displays answer, source page numbers, and supporting context.

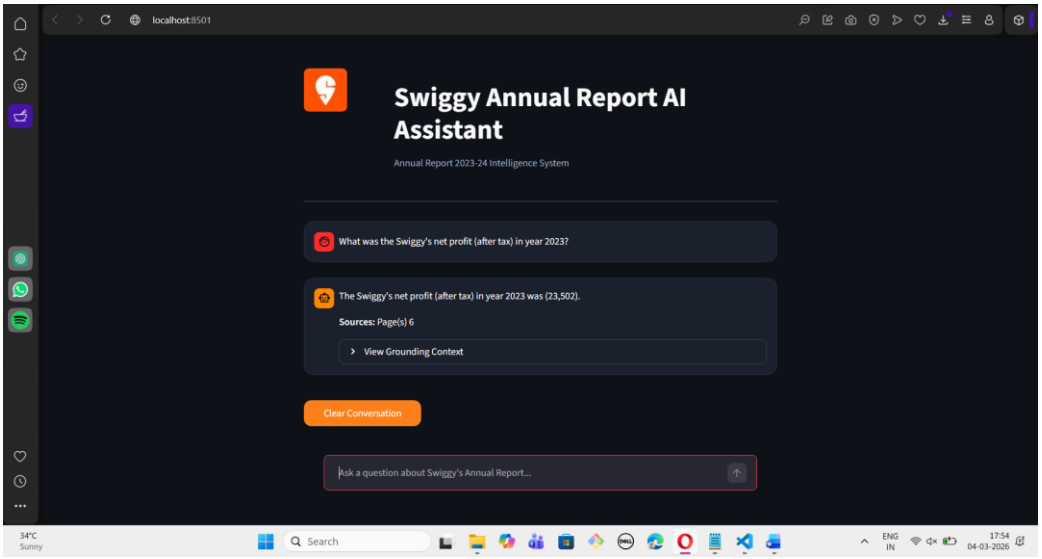
Additional Enhancements

- **Multi-Query Retrieval:** Implemented **to handle non search-friendly queries** by generating multiple query variations to improve retrieval accuracy.
- **Re-Ranking:** Implemented **to filter less relevant retrieved chunks** and select the most relevant context for answer generation.
- **Source Citation:** Displays the **page numbers** from which the answer was derived, to increase reliability.
- **View Context:** Allows users to **view the document context** used to generate the answer.
- **RAG Evaluation:** The system was evaluated using the RAGAS framework, achieving a **Faithfulness score of 1.0** and Answer **Relevancy score of 0.89**.

User Interface

A Streamlit-based chat interface allows users to interact with the system.
Features include:

- Chat-based question input
- AI-generated answers
- Source page citations
- Expandable context view
- Conversation history



Conclusion: This project demonstrates an effective RAG-based AI assistant capable of answering questions from the Swiggy Annual Report.
Techniques such as multi-query retrieval, re-ranking, and source citation improve answer accuracy, transparency, and reliability.